NCCN GUIDELINES FOR PATIENTS®

2025

Rectal Cancer

Presented with support from NCCN NATIONAL COMPREHENSIVE CANCER NETWORK® FOUNDATION Guiding Treatment. Changing Lives.

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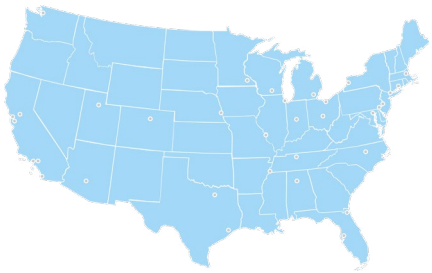
About the NCCN Guidelines for Patients®

National Comprehensive Cancer Network®



Did you know that top cancer centers across the United States work together to improve cancer care? This alliance of leading cancer centers is called the National Comprehensive Cancer Network® (NCCN®).

Cancer care is always changing. NCCN develops evidence-based cancer care recommendations used by health care providers worldwide. These frequently updated recommendations are the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®). The NCCN Guidelines for Patients plainly explain these expert recommendations for people with cancer and caregivers.

These NCCN Guidelines for Patients are based on the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Rectal Cancer, Version 3.2025 — August 26, 2025.

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Fight Colorectal Cancer (Fight CRC) is a leading patient-empowerment and advocacy organization providing balanced and objective information on colon and rectal cancer research, treatment, and policy. We are relentless champions of hope, focused on funding promising, high-impact research endeavors while equipping advocates to influence legislation and policy for the collective good. Get fast answers from ChatCRC, the first and only AI chatbot for colorectal cancer by texting (318) 242-8272 or visiting chatbot.fightcrc.org. For ways to join the fight, visit FightCRC.org.

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NCCN Foundation seeks to support the millions of patients and their families affected by a cancer diagnosis by funding and distributing NCCN Guidelines for Patients. NCCN Foundation is also committed to advancing cancer treatment by funding the nation's promising doctors at the center of innovation in cancer research. For more details and the full library of patient and caregiver resources, visit NCCN.org/patients.

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About rectal cancer

5 What is the rectum?

6 What are polyps?

7 How can you get the best care?

Rectal cancer is common and treatable. This guide describes treatment for adenocarcinoma, the most common type of rectal cancer.

The term colorectal cancer is used to describe cancers that form in either the colon or rectum. While these cancers are similar, their treatment is different. The focus of this guide is rectal cancer.

For information on colon cancer treatment, see the *NCCN Guidelines for Patients: Colon Cancer* at NCCN.org/patientguidelines and on the NCCN Patient Guides for Cancer app.

What is the rectum?

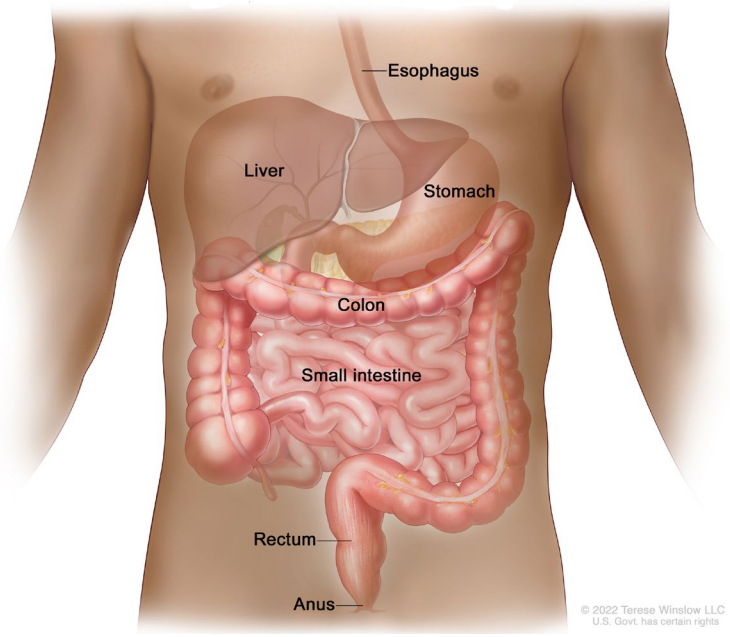
The rectum is the last several inches of the large intestine (bowel). After digested food becomes stool in the colon, it is held in the rectum until it exits the body through the anus.

The rectum is made of layers of tissue. Cancer starts in the innermost layer that comes in contact with stool. If left untreated, cancer cells can grow into the deeper layers of the rectal wall and invade other parts of the body.

Cancer cells can also break off from the rectal tumor and travel through lymph or blood to nearby lymph nodes or to distant organs like the liver or the lungs.

The rectum

The rectum is about 5 inches (12 centimeters) long and forms the last part of the large bowel. Stool is held in the rectum until it leaves the body through the anus.



What are polyps?

Polyps are non-cancerous growths that form on the innermost lining of the intestine.

The most common type of polyp is called an adenoma. While it may take many years, adenomas can become invasive rectal cancer, called an adenocarcinoma. It's the most common type of rectal cancer.

Polyps that rarely turn into cancer include hyperplastic and inflammatory polyps.

Removing polyps can prevent cancer before it starts. Most can be removed during a colonoscopy using a minor endoscopic procedure called a polypectomy. Polyps can also be tested to make sure that cancer hasn't already started to form.

For more information on finding polyps early, see the *NCCN Guidelines for Patients: Colorectal Cancer Screening* at NCCN.org/patientguidelines and on the NCCN Patient Guides for Cancer app.

What's in this resource?

- Testing the tumor for an important feature called mismatch repair deficiency (dMMR)
- Developing a care plan based on the cancer's mismatch repair (MMR) status
- Treatment for polyps with cancer and all stages of rectal cancer
- Survivorship

How can you get the best care?

Advocate for yourself. You're more likely to get the care you want by asking questions and making shared decisions with your care team.

The NCCN Guidelines for Patients will help you understand cancer care. With better understanding, you'll be more prepared to discuss your care with your team and share your concerns. Many people feel more satisfied when they play an active role in their care.

You may not know what to ask your care team. That's common. Each chapter in this book ends with an important section called "Questions to ask." These suggested questions will help you get more information on all aspects of your care.

Why you should read this book

Making decisions about cancer care can be stressful. You may need to make tough decisions under pressure about complex choices.

The NCCN Guidelines for Patients are trusted by patients and providers. They clearly explain current care recommendations made by respected experts in the field. Recommendations are based on the latest research and practices at leading cancer centers.

Cancer care is not the same for everyone. By following expert recommendations for your situation, you are more likely to improve your care and have better outcomes as a result. Use this book as your guide to find the information you need to make important decisions.

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16 Questions to ask

This chapter discusses testing and other steps needed to create your treatment plan. All rectal cancers should be tested for mutations (changes) in genes that fix damaged DNA, called mismatch repair (MMR) genes.

Colonoscopy

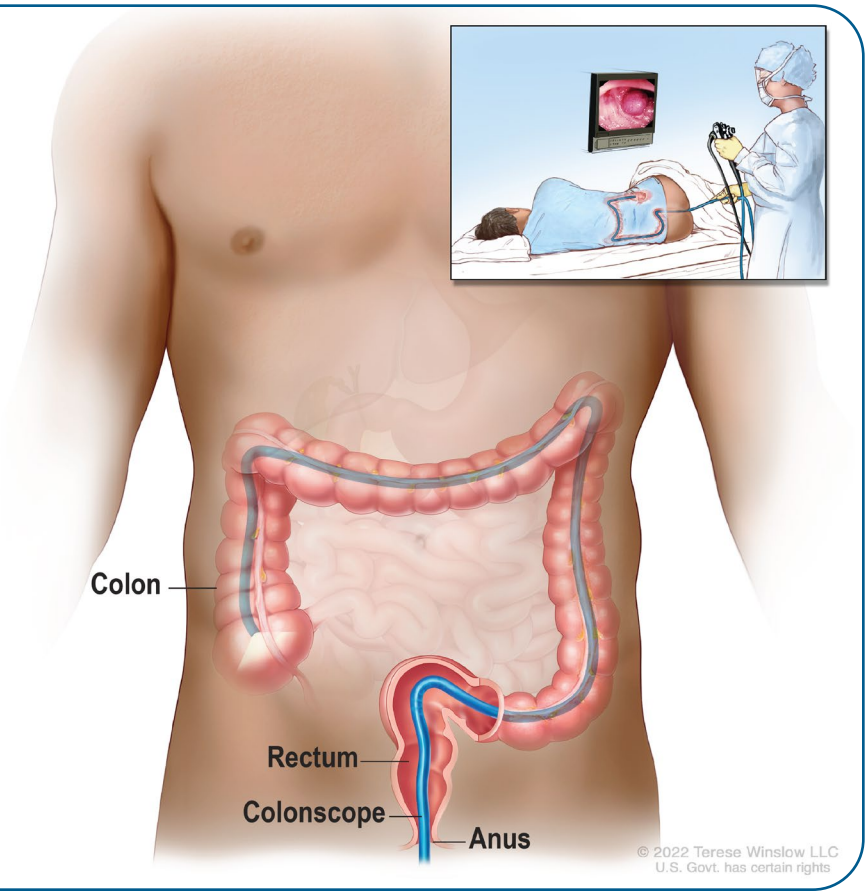
Most colorectal cancers are found during colonoscopy. This procedure allows a health care provider, usually a gastroenterologist, to examine the entire large bowel.

If they see any polyps or lesions, they will remove them during the procedure using a cutting tool. This is called a polypectomy. An expert called a pathologist will look for cancer cells in the polyp(s) using a microscope. If any are found, they will make a cancer diagnosis.

The diagnosis and other findings will be included in a document called a pathology report. Ask your provider for a copy.

Colonoscopy

Colonoscopy is a procedure that allows your doctor to see and remove any abnormal tissue from the large intestine. A thin device is inserted through the anus, up the rectum, and into the colon. The device has a light, a camera, and a cutting tool.



It's common to have another colonoscopy after rectal cancer is diagnosed to thoroughly examine the large bowel and look for other possible areas of cancer. This is called a diagnostic colonoscopy.

For more information on screening colonoscopy and what to expect, see the *NCCN Guidelines for Patients: Colorectal Cancer Screening* at NCCN.org/patientguidelines and on the NCCN Patient Guides for Cancer app.

Mismatch repair testing

All rectal cancers should be tested for mutations (changes) in genes that fix damaged DNA, called mismatch repair (MMR) genes. Testing involves analyzing a piece of the rectal tumor in a lab.

Depending on the method used, an abnormal result is called either mismatch repair deficiency (dMMR) or microsatellite instability-high (MSI-H). About 1 in 6 colorectal cancers have one of these biomarkers. Biomarkers are features of a cancer that can help guide your treatment.

Tumors that don't have either of these changes are described as mismatch repair proficient (pMMR) or microsatellite stable (MSS).

Guide 1
Testing and care for rectal cancer

Necessary tests	• Colonoscopy • Biopsy • Testing the tumor for mismatch repair deficiency (dMMR) • Blood tests: Complete blood count, comprehensive metabolic panel, carcinoembryonic antigen • MRI of your pelvis (to see the extent of cancer in and near the rectum) • CT of your chest and abdomen (to check for tumors in the liver and lungs)
Other tests or care you may receive	• Endorectal ultrasound (if you can't have MRI) • *PIK3CA* testing • *DPYD* testing • Help to quit smoking • Family planning and options for preserving fertility

Family health history

Your provider will ask about the health history of your biological (related by blood) family members. Those with a first-degree relative with colorectal cancer are more likely to have this cancer compared to those without a family history of colorectal cancer.

Most colorectal cancers occur for unknown reasons. While rare, some people are born with a disorder that makes them more likely to get colorectal and other cancers. These are called inherited cancer syndromes. Two such conditions include Lynch syndrome and familial adenomatous polyposis (FAP).

People born with Lynch syndrome are at high risk of developing colorectal, endometrial, and ovarian cancers. Lynch syndrome is caused by inherited mutations of the MMR genes. For cancers found to have the dMMR or MSI-H biomarker, testing for Lynch syndrome is recommended.

FAP causes many polyps to form in the colon and rectum. The polyps start as benign growths, but can become invasive colorectal cancer over time. Cancer often develops by age 50 in people with FAP.

Genetic testing

If an inherited cancer or cancer syndrome is suspected, your provider may refer you to a genetic counselor. This expert can speak with you and your family about testing for syndromes related to colorectal cancer.

To be tested, you must provide a sample of blood or saliva. A pathologist tests the sample for gene mutations that cause these syndromes. Speaking with a genetic counselor is recommended for any positive genetic testing result.

PIK3CA testing

PIK3CA is a gene involved in normal cell growth. At some point after being conceived, a mutation (change) can occur in this gene that increases your risk of colorectal and other cancers. Gene changes that happen after conception are called **somatic mutations**. Gene differences you are born with are called **germline variants**.

Cancers with a *PIK3CA* mutation tend to grow quickly and return after treatment. Before surgery, if the cancer is thought to be stage 2 or 3, your provider may order *PIK3CA* testing. The results can help guide decisions about taking aspirin after treatment. Aspirin may work better at preventing recurrence (the return of cancer) in cancers with this biomarker.

DPYD testing

Capecitabine and fluorouracil (5-FU) are chemotherapy medicines called fluoropyrimidines. They are often used to treat colorectal cancer. Your body uses a special enzyme (protein) called DPD to break down these medicines so they can work properly and safely.

Around 1 in 15 people inherit a difference in the gene involved in making DPD, the *DPYD* gene. This is called DPD deficiency. In these people, fluoropyrimidines can build up and cause severe and sometimes life-threating

side effects, referred to as fluoropyrimidine toxicity.

Signs and symptoms include mouth sores, abdominal pain, blood in the stool, nausea, vomiting, and diarrhea. The skin on the palms of your hands and soles of your feet may start peeling and become red, swollen, or numb. Other possible side effects include shortness of breath and hair loss.

If you develop any of the symptoms mentioned above in the days after receiving capecitabine or 5-FU, contact your care team right away. There is a U.S. Food and Drug Administration (FDA)-approved treatment for this uncommon side effect of fluoropyrimidines.

If treatment with a fluoropyrimidine is planned, talk to your provider about fluoropyrimidine toxicity. After understanding your risk, you may choose to have genetic testing to learn if you have the gene variant that causes DPD deficiency.

Blood tests

CEA blood test

Carcinoembryonic antigen (CEA) is a protein found in blood. People with colorectal cancer tend to have more of this protein than normal.

Monitoring CEA can be helpful for some cancers that are only in the rectum. If the level rises, it could signal that the cancer has spread. If the level is high at diagnosis, it could suggest that the cancer has already spread.

ctDNA Testing

Also called a liquid biopsy, circulating tumor DNA (ctDNA) testing looks for small pieces of DNA released by tumor cells into the blood. It can detect microscopic disease that may remain in the body after rectal surgery. **At this time, ctDNA testing is not recommended as part of standard care for colorectal cancer.**

Monitoring CEA isn't helpful for everyone. Pregnant people and tobacco users may also have more CEA than average in their blood.

Complete blood count

Cancer and other health problems can cause low or high blood cell counts. A complete blood count (CBC) measures the number of white blood cells, red blood cells, and platelets in a blood sample.

White blood cells help you fight infection. Red blood cells carry oxygen throughout the body. Platelets help wounds heal by forming blood clots.

Chemistry profile

A chemistry profile (also called a comprehensive metabolic panel) is a group of tests that provides information about how well your kidneys, liver, and other organs are working.

Imaging

Imaging tests can show areas of cancer inside the body. This information helps your care team plan treatment. A radiologist will interpret and convey the imaging results to your oncologist.

Your care team will tell you how to prepare for these tests. You may need to stop taking some medicines and stop eating and drinking for a few hours beforehand.

Some imaging tests use contrast. Contrast is a substance that makes the pictures clearer. It is injected into your vein and/or mixed with a liquid to drink. Tell your doctor if you've had problems with contrast in the past. You may be given or advised to take a steroid and an antihistamine medication (like Benadryl) before the scan. It will lower the risk of an allergic reaction.

MRI

If surgery is needed or being considered, your health care provider will order magnetic resonance imaging (MRI) of your pelvis.

MRI can show how deep into the rectal wall the cancer has grown and whether it has spread to nearby lymph nodes. Special techniques called a staging protocol are used to stage the cancer. Contrast may or may not be used. If needed, MRI may also be used to look for cancer spread in the abdomen, especially in the liver.

MRI doesn't use radiation and is safe for most people. People with certain heart monitors, pacemakers, or some types of surgical clips generally can't have an MRI. Tell your team if you get nervous in small spaces. You may be given a sedative (medicine) to help you relax.

A gel may be inserted into the rectum before MRI imaging. A light coil device or blanket will

MRI

An MRI can show how deep into the rectal wall the cancer has grown and whether it has spread to nearby lymph nodes.

be placed on top of you. It sends and receives radio waves to record the images. The device covers your body from below your chest to the top of your legs. Straps may be used to help you stay in place. You may feel a bit warm during the scan.

Endorectal ultrasound

While a pelvic MRI is preferred for imaging rectal cancer, an endorectal ultrasound (EUS) is sometimes performed instead.

Using sound waves, this test can also show the extent of cancer in the pelvis. A small probe is inserted into the rectum. Echoes form a picture that can be seen by your doctor on a screen. If needed, EUS may also be used to guide a biopsy of lymph nodes or other areas near the rectum.

CT

A computed tomography (CT) scan of your chest and abdomen is recommended as part of initial testing. CT can show if the cancer has spread to the liver, lungs, or other organs. Contrast will be used.

During the scan you will lie face-up on a table that moves through a tunnel-like machine. A technician will be nearby. You will be able to hear and talk to them at all times.

Team approach

When possible, treatment for rectal cancer is best delivered by a team of experts in the following specialties:

- digestive health (gastroenterology)
- chemotherapy (medical oncology)
- surgery (surgical oncology)
- radiotherapy (radiation oncology)
- imaging (radiology)

Fertility and family planning

For unknown reasons, colorectal cancer is being diagnosed more often in young adults. Treatment for rectal cancer, especially radiation therapy, can impair your ability to have children.

Radiation damages the ovaries and testicles causes them to stop making hormones needed for natural pregnancy. Damage to the ovaries is known as premature ovarian insufficiency (POI) or radiation-induced premature menopause. See *Chapter 6: Survivorship* for more information on this side effect.

In some cases, treatment with chemotherapy can also impair fertility by damaging sperm or eggs.

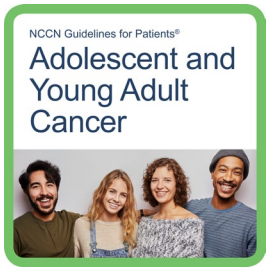
If you want the option of having children after treatment or are unsure, ask your care provider about any fertility-related risks of your treatment. They may refer you to an expert to learn about options for preserving your fertility, including those described next.

Sperm banking

Sperm banking stores semen for later use by freezing it in liquid nitrogen. The medical term for this is semen cryopreservation.

Egg freezing

Like sperm banking, unfertilized eggs can be removed, frozen, and stored for later use. The medical term for this is oocyte cryopreservation.



For more information on fertility and family planning, see the NCCN Guidelines for Patients: Adolescent and Young Adult Cancer at NCCN.org/patientguidelines and on the NCCN NCCN Patient Guides for Cancer app.

Ovarian tissue banking

This method involves removing part or all of an ovary and freezing the part that contains the eggs. The frozen tissue that contains the eggs can later be unfrozen and put back in the body.

Ovarian transposition

This procedure moves one or both ovaries out of the range of the radiation beam. The medical name for this procedure is oophoropexy.

Key points

- All rectal tumors should be tested for mismatch repair deficiency (dMMR) or microsatellite instability-high (MSI-H). Cancers that don't have either of these biomarkers are described as mismatch repair proficient (pMMR) or microsatellite stable (MSS).
- If the cancer is found to have the dMMR or MSI-H biomarker, genetic testing (of your blood or saliva) for Lynch syndrome is recommended.
- MRI of the pelvis using a rectal cancer staging technique can show the extent of the cancer in the rectum and nearby lymph nodes.
- CT of the chest and abdomen can show if the cancer has spread to the liver, lungs, or other organs.
- For cancers thought to be stage 2 or 3 before surgery, *PIK3CA* testing can help guide decisions about taking aspirin after treatment.
- After understanding your risk for fluoropyrimidine toxicity, you may choose to have testing for gene variants that cause DPD deficiency.
- If you want the option of having children in the future or are unsure, ask your care team about any fertility-related risks of your treatment plan and options for preserving your fertility.

Questions to ask

- Does the cancer have the mismatch repair deficiency (dMMR) biomarker?
- What if I am pregnant, or planning to become pregnant in the future?
- Colorectal cancer runs in my family. What does this mean for my care?
- My doctor said I need fluoropyrimidine chemotherapy. Is it safe and should *DPYD* testing be considered?
- Is it worth it to go through treatment?

3
Treatment for early cancer

18 Polyps with cancer

19 Endoscopic submucosal dissection

20 Transanal local excision

21 Abdominal surgery

24 After abdominal surgery

25 Key points

25 Questions to ask

This chapter explains treatment for polyps with cancer and for smaller rectal tumors.

Polyps with cancer

There are 2 main shapes of polyps. Pedunculated polyps are shaped like mushrooms and stick out from the colon wall. They have a stalk and round top.

Sessile polyps are flatter and don't have a stalk. A polyp in which cancer has just started to grow is called a malignant (cancerous) polyp.

Most polyps can be removed during a colonoscopy, using a minor endoscopic procedure called a polypectomy. Often, no further treatment is needed.

In other cases, resection (surgical removal) of a larger piece of the rectum is needed. This depends on the following factors:

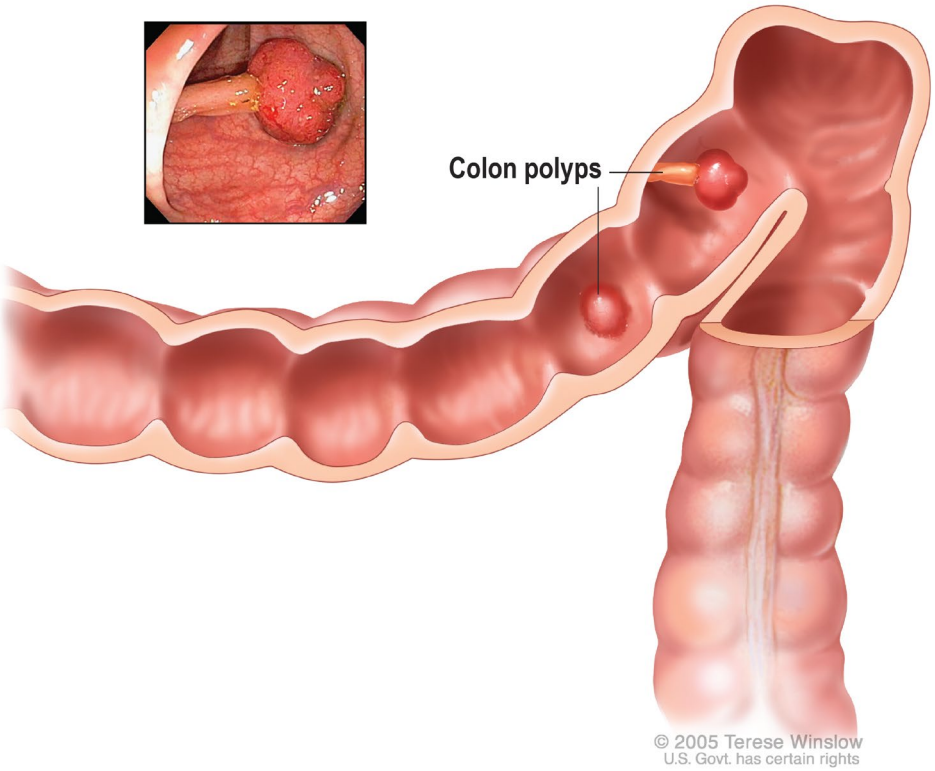
- The size and shape of the polyp
- The polypectomy results
- The results of testing the removed tissue

Before deciding whether resection is needed after a polypectomy, your provider will review the results of testing with you and discuss your options.

Sessile polyps are more likely to return after treatment. Imaging with endoscopic ultrasound

Polyps

Polyps are growths on the lining of the bowel wall. Pedunculated polyps look like mushrooms and stick out from the colon wall. Sessile polyps are flatter and don't have a stalk.



(EUS) or pelvic magnetic resonance imaging (MRI) is recommended for these polyps to help guide decisions about surgery.

If cancer that formed in a polyp isn't found early enough to be removed by polypectomy, surgery is needed. Surgery is also needed for rectal cancer that forms as a lesion on the rectal wall.

The true extent of the cancer can't be known until after surgery. But it can be estimated based on certain test results. This best guess is called the clinical or pre-surgery stage. It is used to determine your treatment options.

Endoscopic submucosal dissection

If testing finds that the tumor hasn't grown beyond the second (muscle) layer of the rectal wall, it is a **T1 tumor**.

Endoscopic submucosal dissection (ESD) is a recommended option for all T1 tumors. This minimally invasive procedure removes small rectal cancers in one piece, often curing them. This procedure is sometimes called endoscopic mucosal resection (EMR).

First, an endoscope is used to locate the tumor in the rectal wall. A tool is guided through the scope to inject fluid between the tumor and the layer of muscle beneath it. This helps lift and separate the tumor, making it easier to cut off.

Most people don't need more treatment after ESD. **See Guide 2** for recommendations.

Guide 2
Surveillance after endoscopic submucosal dissection (ESD)

Length of surveillance	5 years
Flexible sigmoidoscopy	Every 6 months. The first may be as soon as 3 months after ESD.
Endorectal ultrasound or pelvic MRI with contrast	Years 1 and 2: Every 3 to 6 months Years 3, 4, and 5: Every 6 months

Transanal local excision

For **T1 tumors** located at the end of the rectum, removing them through the anus may be an option. This is called transanal local excision.

The surgeon cuts through all layers of the rectal wall to remove the cancer and some surrounding normal rectal tissue. Lymph nodes are not removed.

If surgery confirms that the tumor is T1 and low-risk, treatment is finished. See **Guide 3** for surveillance recommendations.

Advanced techniques for removing tumors higher in the rectum include transanal endoscopic microsurgery (TEM) and transanal minimally invasive surgery (TAMIS).

When more treatment is needed

If surgery finds that the tumor has high-risk features, or is larger or deeper than expected, the **preferred next treatment** is abdominal surgery. See the next page for details.

Other options include chemoradiation and short-course radiation therapy. Chemotherapy may follow whichever of these is planned. Surgery is often the next step. Or, if there are no signs of cancer, closely monitoring the cancer may be an option.

Guide 3
Surveillance after transanal local excision

Length of surveillance	5 years
Proctoscopy	Years 1 and 2: Every 3 to 6 months Years 3, 4, and 5: Every 6 months
Endorectal ultrasound or pelvic MRI with contrast	Years 1 and 2: Every 3 to 6 months Years 3, 4, and 5: Every 6 months
Colonoscopy	One year after surgery. The timing of your next will depend on the results.

Abdominal surgery

Removing the cancer through the abdomen is recommended for tumors that have invaded the muscle layer of the rectal wall. In staging terms, these are **T2 tumors**.

Abdominal surgery is also a recommended option for:

- smaller (T1) tumors, and
- some low-risk, larger (T3) tumors that haven't spread to lymph nodes.

This method involves cutting into the abdomen to reach and remove the rectal tumor, some surrounding healthy tissue, and nearby lymph nodes.

The tissue and lymph nodes removed during surgery are sent to a lab for testing. A pathologist assesses how far the cancer has grown within the rectal wall and whether any removed lymph nodes have cancer.

Using this information, they assign the cancer a stage of I (1), II (2), or III (3). This process is called surgical (or pathologic) staging. The stage is used to determine whether more treatment is needed.

A **stage 1** cancer has grown into the second or third (muscle) layer of the rectal wall. No further treatment is needed.

A **stage 2** cancer has grown through the rectal wall and possibly into nearby tissues or organs. More treatment is almost always needed.

We want your feedback!

Our goal is to provide helpful and easy-to-understand information on cancer. Take our survey to let us know what we got right and what we could do better.
NCCN.org/patients/feedback

A **stage 3** cancer has spread to nearby lymph nodes. More treatment is almost always needed.

The following types of abdominal surgery are described on the next pages:

- Low anterior resection (LAR)
- Abdominoperineal resection (APR)

The type performed depends on the location of the tumor and the extent of the cancer. At least 12 lymph nodes should be removed during all types of transabdominal surgery.

Low anterior resection

A low anterior resection (LAR) is a type of transabdominal surgery used for tumors in the middle or upper rectum.

In addition to the rectal tumor, part or all of the last section of the colon is removed. When possible, the colon is connected to the remaining rectum. This is called a colorectal anastomosis (reconnection).

Otherwise the colon may be attached directly to the anus. This is called a coloanal anastomosis. This allows for the possibility of having near-normal bowel movements.

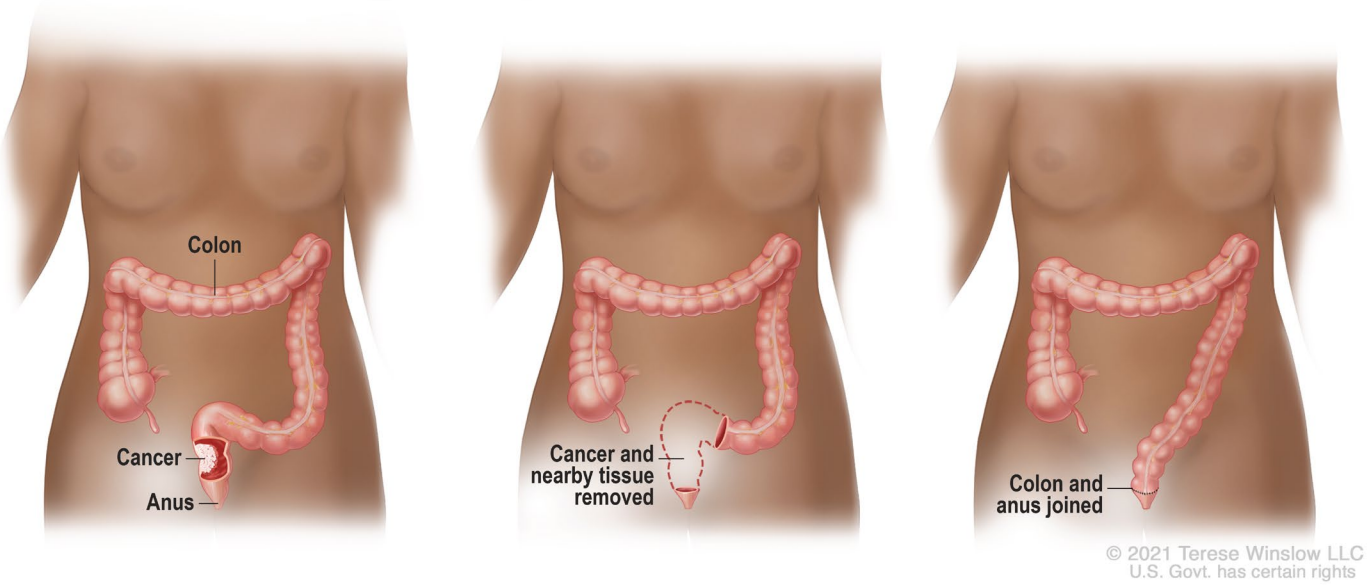
Your surgeon is likely to delay reattaching the bowel in order to let the surgical site heal. In this case, or if the colon can't be connected to the anus, a colostomy is made.

A colostomy connects a part of the colon to the outside of the abdomen. This creates an opening (a stoma) on the surface of the abdomen. Stool exits the body through the stoma and goes into a bag attached to the skin.

A colostomy may only be needed for a short time to allow the rectum to heal before being reconnected to the colon. But sometimes it is needed permanently.

Low anterior resection (LAR)

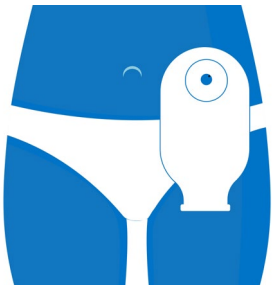
LAR is a type of abdominal surgery used to remove tumors in the mid to upper rectum. The rectal tumor and part or all of the sigmoid colon is removed. When possible, the colon is connected to the remaining rectum.



Abdominoperineal resection

This type of transabdominal surgery is used for tumors in the **lower rectum**. These tumors may have grown into the anus or nearby pelvic floor muscle.

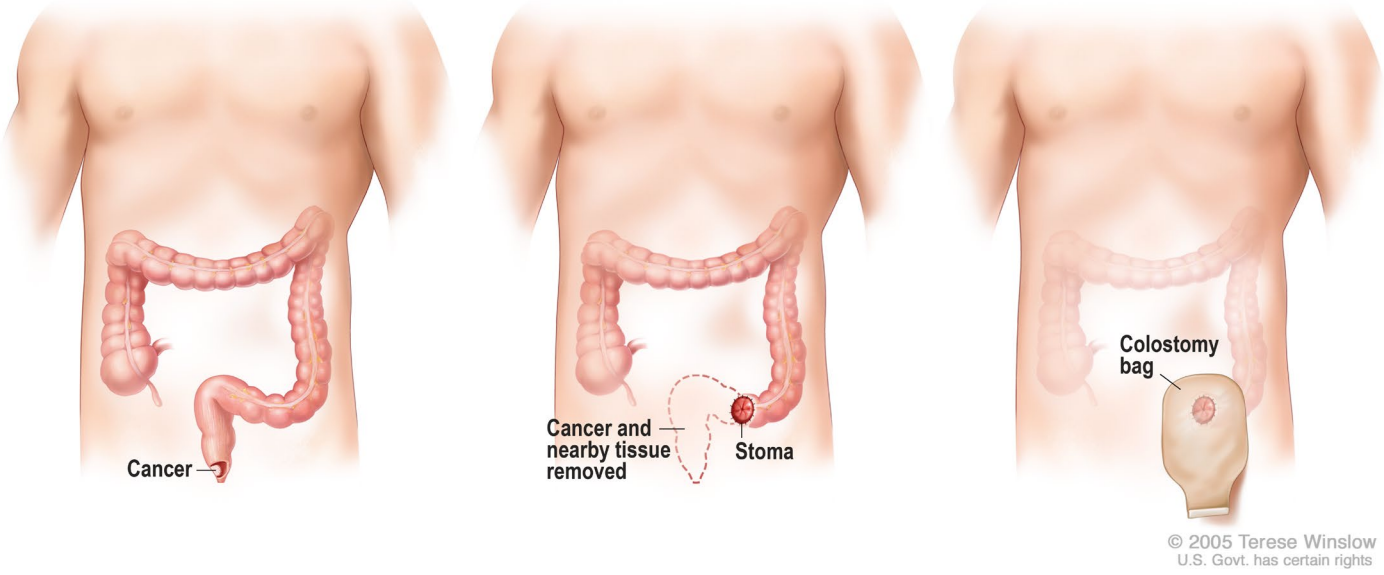
Abdominoperineal resection (APR) typically involves removing the anus, the rectum, the area where the rectum and colon meet, and other tissue. In some cases, the levator muscles are also removed. It may be possible to keep the outer ring of the anus. A permanent colostomy is always needed.



A permanent colostomy is always needed after APR surgery. Consider joining an ostomy support group or seeing a health care provider that specializes in ostomy care. See *Chapter 6: Survivorship* for more information.

Abdominoperineal resection (APR)

APR is used for tumors in the lower rectum that may have grown into the anus or nearby muscle. It involves removing the rectum and anus and creating a permanent colostomy for stool to leave the body.



After abdominal surgery

Sometimes abdominal surgery to remove a small (T1 or T2) tumor finds that the cancer is larger or deeper than expected. In this case, chemotherapy **and** long-course chemoradiation typically come next.

The order in which you receive these treatments can vary. Your care team will determine the best sequence for you. Sometimes it's possible to safely skip chemoradiation.

More information on these treatments is provided next.

Chemotherapy

Chemotherapy given after surgery to kill remaining cancer cells is described as adjuvant chemotherapy. Either FOLFOX or CAPEOX is recommended. The FOLFOX regimen includes the following medicines:

- Folinic acid (leucovorin calcium)
- Fluorouracil (5-FU)
- Oxaliplatin

The CAPEOX regimen includes:

- Capecitabine
- Oxaliplatin

Guide 4
Surveillance after abdominal surgery

Stage 1 (determined by surgical staging)	**Colonoscopy**	One year after surgery. The timing of your next will depend on the results. Other testing is only needed if you develop symptoms or if your provider suspects recurrence.
Stages 2 and 3 (determined by surgical staging)	**Physical exam and CEA blood test**	Years 1 and 2: Every 3 to 6 months Years 3, 4, and 5: Every 6 months
	CT of your chest, abdomen, and pelvis	Every 6 to 12 months for 5 years
	Colonoscopy	**If you didn't have a complete colonoscopy at diagnosis,** colonoscopy is recommended 3 to 6 months after surgery. **If you had a complete colonoscopy at diagnosis,** colonoscopy is recommended 1 year after surgery. If no advanced adenomas are found, repeat in 3 years. After that, repeat every 5 years.

Oxaliplatin can cause nerve damage (called neuropathy) to your fingers and toes. Symptoms include numbness, cramping, tingling, or pain in these areas.

Chemotherapy is given in cycles of treatment days followed by days of rest. This allows your body to recover between cycles. Cycles vary in length depending on which drugs are used.

Long-course chemoradiation

Chemoradiation is used most often to treat locally advanced cancers. For details, see *Chapter 4: Treatment for locally advanced rectal cancer*.

Surveillance

For cancers removed by abdominal surgery, recommended surveillance testing is shown in **Guide 4** on the previous page.

For stage 2 and 3 cancers with a *PIK3CA* mutation, taking aspirin for 3 years after surgery is recommended. In cancers with this biomarker, aspirin may work better at preventing recurrence than in cancers without it.

In addition to surveillance testing, a range of other care is important for cancer survivors. See the *Survivorship* chapter of this resource for more information.

Key points

- Sometimes polypectomy is the only treatment needed for a polyp that contains cancer. In other cases, surgery may be needed to remove a larger piece of the rectum.
- Surgery is often recommended for early rectal cancers when possible. The type of surgery performed depends on the location and extent of the tumor.
- Treatment options may include endoscopic submucosal dissection (ESD), transanal local excision, and transabdominal surgery.
- Types of transabdominal surgery include low anterior resection (LAR) and abdominoperineal resection (APR).
- For small tumors found to be more advanced after abdominal surgery, treatment with both long-course chemoradiation and chemotherapy typically follows.
- For stage 2 and 3 cancers with a *PIK3CA* mutation, taking aspirin for 3 years is recommended after surgery.

Questions to ask

- Do I need more treatment than a polypectomy?
- If I need surgery, what type do you recommend?
- When treatment is over, how often should I get checked for new polyps?
- Is recurrence likely after treatment for small rectal tumors?
- I'm not currently taking aspirin and my cancer doesn’t have a *PIK3CA* mutation. Should I start taking aspirin? Will it hurt me?

4
Treatment for locally advanced cancer

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This chapter describes treatment for rectal cancers that have spread beyond the rectum but not to distant areas of the body.

What is locally advanced rectal cancer?

The term locally advanced is used to describe rectal cancers that:

- have grown through the rectal wall (a T3 or T4 tumor), or
- have spread to nearby lymph nodes, or
- can't be removed using surgery.

Radiation therapy

Treatment for loclaly advanced rectal cancer often involves external beam radiation therapy (EBRT), chemotherapy, or both.

In EBRT, radiation is delivered using a large machine outside the body. The radiation passes through skin and other tissue to reach the tumor. The radiation beams are shaped to the cancer site. This helps minimize damage to healthy tissue.

There are different types of EBRT. The type used depends on the location and size of the tumor and other factors.

Side effects of radiation therapy

- Feeling tired and worn out
- Hair loss in the treated area
- Changes to urination and bowel movements
- Diarrhea
- Nausea or vomiting
- Late side effects can include infertility, sexual problems, bowel problems, reduced bone density, and second cancers
- Vaginal dilators can help with sexual side effects of radiation therapy – see *Chapter 6: Survivorship* for more information
- Ask your care team for a full list of possible side effects

Long-course chemoradiation

EBRT is often used with chemotherapy to treat rectal cancer. When given with radiation, chemotherapy can make it easier for radiation to kill cancer cells. This is called radiosensitizing chemotherapy.

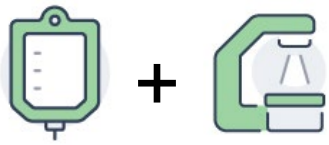
In long-course chemoradiation. radiation therapy is given in 25 to 28 treatment sessions, called fractions. Most people have 5 sessions per week. Chemotherapy is given during the same 5- to 6-week period.

Either 5-fluorouracil or capecitabine is usually given for chemotherapy. 5-FU is given using an infusion pump hooked into a central access device like a portacath. You take this pump home with you for up to a week at a time. Capecitabine is a pill that you take twice daily on radiation days.

Short-course radiation therapy

Another method of radiation treatment for rectal cancer is short-course radiation therapy. This method delivers a higher dose of radiation over a much shorter time period, typically in 5 treatment sessions. Chemotherapy isn't given.

Long-course chemoradiation



- 25 to 28 radiation treatment sessions (Monday through Friday)
- Chemotherapy is given during the same time period
- Either 5-fluorouracil or capecitabine is usually given for chemotherapy

Short-course radiation therapy

- High-dose radiation is given over a short time period
- Typically only 5 treatment sessions are needed

For many years, long-course chemoradiation was the standard approach for treating locally advanced rectal cancer. Short-course radiation therapy is another option found to have similar outcomes in certain cases.

Treatment for pMMR/MSS cancer

Treatment for locally advanced pMMR/MSS cancers typically begins with 12 to 16 weeks of chemotherapy and either long-course chemoradiation or short-course radiation therapy. For most people, transabdominal surgery is the next step. This approach is known **as total neoadjuvant therapy (TNT)**.

Sometimes, chemotherapy and chemoradiation work so well that no cancer can be detected in the body. Doctors call this a complete response. If this happens, research suggests that having surgery doesn't always lead to better outcomes.

You may be given the option to monitor the cancer closely instead. This is only an option for carefully selected patients who agree to close surveillance programs. **See Guide 5** for recommended surveillance in those treated without surgery.

The benefits and risks of taking a watch-and-wait approach instead of having surgery aren't fully known. The decision to have surgery (or not) should be one that you make with your doctor after careful consideration.

Guide 5
Surveillance for those treated without surgery

Physical exam and CEA blood test	The first 2 years: Every 3 to 6 months The following 3 years: Every 6 months
DRE and proctoscopy or flexible sigmoidoscopy	The first 2 years: Every 3 to 4 months The following 3 years: Every 6 months
MRI of your rectum	Every 6 months for up to 3 years
CT of your chest and abdomen	Every 6 to 12 months for 5 years. When MRI imaging stops after 3 years, CT scanning will also include your pelvis.
Colonoscopy	In one year. The timing of your next will depend on the results.

Treatment for dMMR/MSI-H cancer

Immunotherapy

The **preferred** treatment approach for locally advanced cancers with this biomarker begins with immunotherapy. At this time, recommended checkpoint inhibitors include:

- Dostarlimab-gxly (Jemperli)
- Nivolumab (Opdivo)
- Pembrolizumab (Keytruda)

Immunotherapy is recommended for up to 6 months. The extent of the cancer will be checked every 2 to 3 months during this time. For some people, no further treatment will be needed. Surveillance for those treated without surgery will begin. See **Guide 5.**

If cancer remains after 6 months of immunotherapy, treatment with long-course chemoradiation or short-course radiation therapy is recommended. Transabdominal surgery is often the next step. Or, the selected approach may work so well that no cancer can be detected in the body. This is called a **complete response**.

If this happens, you may be able to monitor the cancer closely instead. This is only an option for carefully selected patients who agree to close surveillance programs. See **Guide 5** for recommended surveillance in those treated without surgery.

Immunotherapy side effects

Checkpoint inhibitors have unique side effects. Unlike other cancer treatments, the side effects of immunotherapy occur because the immune system is attacking healthy cells in the body. Learning to recognize possible reactions can help you report them sooner to your care team.

More information is available at NCCN.org/patientguidelines and on the NCCN Patient Guides for Cancer app.

Total neoadjuvant therapy

If you can't have immunotherapy, another recommended approach is called **total neoadjuvant therapy**. Treatment begins with either long-course chemoradiation or short-course radiation therapy. The next step is 12 to 16 weeks of chemotherapy, usually with FOLFOX or CAPEOX. Your provider may also consider the FOLFIRINOX regimen.

After treatment, you will have imaging to see if abdominal surgery is an option. If imaging results show cancer, surgery is recommended. See **Guide 4** for recommended surveillance after abdominal surgery.

For stage 2 and 3 cancers with a *PIK3CA* mutation, taking aspirin for 3 years is recommended after surgery. In cancers with this biomarker, aspirin may work better at preventing recurrence than in cancers without it.

If you aren't a candidate for surgery, chemotherapy is recommended.

Recurrence

If cancer returns to the rectum or anywhere in the pelvis, it's called a **local recurrence**.

Your options for treatment depend on whether the cancer is (or may become) resectable, and whether your pelvis has already been treated with radiation. If you've already had chemoradiation, it's usually not recommended to treat recurrence in the same area.

If the cancer is small enough, your first treatment may be surgery. Chemoradiation is often given next to kill any remaining cancer cells in the pelvis. In other cases, chemotherapy and either long-course chemoradiation or short-course radiation therapy is given first. The goal is to shrink the cancer enough that it can be removed surgically.

If surgery isn't possible, you may have treatment with systemic therapy, chemoradiation, or short-course radiation.

If systemic therapy is planned, your doctor will consider the following when selecting a regimen:

- Therapies you've already received
- Whether the tumor has any biomarkers
- How well you are expected to tolerate certain systemic therapies

When cancer returns to areas far from the rectum, it's called a **distant recurrence**. See the next chapter for information.

Key points

- Locally advanced cancers have grown through the rectal wall or have spread to nearby lymph nodes.
- Treatment for locally advanced mismatch repair proficient (pMMR) cancers starts with chemotherapy and either long-course chemoradiation or short course radiation therapy. Surgery usually follows.
- The preferred treatment for locally advanced mismatch repair deficient (dMMR) cancers is checkpoint inhibitor immunotherapy.
- If cancer remains after 6 months of immunotherapy, treatment with long-course chemoradiation or short-course radiation therapy is recommended. Transabdominal surgery is often the next step.
- If you can't have immunotherapy, a second option for locally advanced dMMR cancers begins with either chemoradiation or radiation therapy, followed by chemotherapy alone. Surgery will follow if needed and possible.

Questions to ask

- Will I have total neoadjuvant therapy?
- What is the simulation session for radiation therapy? What can I expect?
- How likely is it that surgery will be possible after total neoadjuvant therapy?
- What follow-up testing is needed for my cancer stage?
- How likely is the cancer to return after treatment? Can I do anything to lower the risk?

5
Treatment for metastatic cancer

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Rectal cancer spreads most often to the liver and less often to the lungs, abdomen, or other areas. Treatment for metastatic cancer depends on whether the cancer has the mismatch repair deficiency (dMMR) biomarker.

Biomarker testing

Biomarkers are features of a cancer that can help guide your care. They are often mutations (changes) in specific genes. When possible, biomarker testing is performed on a piece of tumor tissue removed during a biopsy or surgery. If this isn't possible, a sample of your blood can be tested instead.

All metastatic rectal cancers should be tested for the following biomarkers:

- Mismatch repair deficiency (dMMR) or microsatellite instability-high (MSI-H), if not yet performed
- *RAS* (*KRAS* and *NRAS*) mutations
- *BRAF* mutations
- HER2 amplification

Testing for many biomarkers at one time is called next-generation sequencing (NGS). NGS can find other, rare biomarkers for which targeted treatments may be available, such as *POLE/POLD1* mutations, *RET* fusions, and *NTRK* fusions.

Supportive care is available for everyone with cancer. It isn't meant to treat the cancer, but rather to help with symptoms and make you more comfortable.

Local therapies

Your treatment options may include local therapies. These are treatments that target liver and lung tumors directly.

Some are interventional oncology/radiology techniques, also known as image-guided therapies. These techniques use imaging, such as ultrasound, computed tomography (CT), or PET/CT to deliver minimally invasive cancer treatments. Using imaging during the procedure allows your doctor to precisely target the tumor(s) and confirm treatment success.

A team of experts can determine the best local therapy for your metastatic tumor(s). To learn if surgery or treatment with other local therapies is an option, your case should be evaluated by a multidisciplinary team of experts. The team should include a surgeon experienced in removing liver and lung tumors and an interventional oncologist/radiologist.

Resection

Surgery to remove liver or lung tumors is called resection. If surgery isn't expected to completely remove the tumors, image-guided thermal ablation (described next) may be paired with surgery. And, if the tumors are small and can be completely destroyed, ablation can be used in place of surgery.

If a liver resection is needed, your liver may need to be enlarged first. This is done using a minimally invasive procedure called portal vein embolization. An interventional radiologist inserts a catheter into certain veins in the liver. This blocks the blood vessel to the liver tumor, causing the healthy part of the liver to grow.

Tumor ablation

Image-guided thermal ablation is the use of thermal energy to destroy small liver or lung tumors. A special needle, called a probe, is placed into or next to the tumor. The probe delivers energy to the tumor while minimizing damage to surrounding normal tissue.

Radiofrequency ablation and microwave ablation are commonly used thermal methods that kill cancer cells using heat. Cold energy (cryoablation) is sometimes used, mostly for lung tumors. Less commonly, your provider may recommend ablation that uses electrical pulses (irreversible electroporation) for tumors located in areas where resection or thermal ablation can't be used.

Thermal ablation, like microwave or radiofrequency ablation, may be used with surgery, or alone for small tumors that can be fully removed or destroyed. It will only be used if all visible areas of cancer can be destroyed. Ablation may be performed by either an interventional radiologist or a surgeon. For small tumors, ablation can be done in a single session in the interventional radiology department.

Intra-arterial therapies

If the cancer has spread only (or mainly) to the liver, treatment with intra-arterial liver-directed therapies may be an option. These therapies may be considered for liver tumors that:

- Didn't improve (or stopped improving) with chemotherapy, and
- Cannot be resected or ablated.

Intra-arterial therapies treat liver tumors with chemotherapy beads (chemoembolization) or radioactive spheres (radioembolization). If radiation spheres are used, it is known as selective internal radiation therapy (SIRT) or transarterial hepatic radioembolization (TARE). These procedures are performed by interventional oncologists/radiologists.

A catheter is inserted into an artery in your leg or wrist and guided to the liver tumor(s). Once in place, the spheres or beads are injected into the blood vessel leading to the tumor. They collect inside or around the tumor and deliver radiation or chemotherapy, causing the cancer cells to die. The chemotherapy beads can also work to starve the tumor by stopping its blood supply.

The chemotherapy beads are larger than the radiation spheres, which means that they block the small liver arteries where they are placed. This starves the metastatic tumor by stopping its blood supply. The chemotherapy or radiation further damage the cancer cells and cause the tumor to shrink.

In the rare cases where chemotherapy and radiation can't be used, beads that don't contain chemotherapy may be used to physically block blood supply to the tumor. This is called bland embolization.

HAIC

Hepatic arterial infusion chemotherapy (HAIC) is chemotherapy given directly to the liver. It is often given in addition to standard intravenous chemotherapy. Using a port or pump that is usually placed during surgery to remove liver tumors, the drugs are funneled directly into the artery leading to the liver. HAIC should only be performed by medical oncologists at treatment centers with experience in this method.

Stereotactic body radiation therapy

Stereotactic body radiation therapy (SBRT) is a highly specialized type of external radiation therapy. It may be used to treat rectal cancer that has spread to the liver or lungs that can't be resected or ablated.

In SBRT, high doses of radiation are delivered to metastatic tumor(s) using very precise beams. The radiation comes from a large machine outside the body. The radiation passes through skin and other tissue to reach the tumor(s). Treatment with SBRT is typically complete in 5 or fewer sessions, called fractions.

MRI-Linac

The MRI-Linac is a recent development in radiation therapy. This device combines magnetic resonance imaging (MRI) with a radiation machine called a linear accelerator (LINAC).

If the cancer spreads to an area where the tumor moves when you breathe, the MRI-Linac may be helpful. It can “track” and target the tumor in real time while you hold your breath at regular intervals during the treatment.

SBRT for metastatic tumors

A highly specialized type of radiation therapy called SBRT may be used to treat rectal cancer that has spread to the liver, lungs, or bone. Treatment typically takes 5 or fewer sessions.

Stage 4 cancer in the liver or lungs

If the rectal tumor can be safely removed using surgery and the liver or lung tumor(s) can be safely and completely removed or destroyed, the cancer is considered **resectable**.

Resectable pMMR/MSS cancer

At this time, the **preferred** treatment approach for resectable pMMR/MSS cancers begins with chemotherapy. Either FOLFOX or CAPEOX is preferred. After chemotherapy, most people need either short-course radiation therapy or chemoradiation next.

After chemotherapy and any radiation therapy, the next step is surgery to remove the rectal tumor. The liver and lung tumors may be resected, ablated, or destroyed using other local therapies at this time, or at a later date.

Another recommended approach for resectable stage 4 cancers is to have short-course radiation or chemoradiation first, followed by chemotherapy and then surgery.

Treatment with chemotherapy and radiation works so well for some people that they are able to avoid surgery. **See Guide 5 on page 29** for recommended surveillance testing for those treated without surgery.

Unresectable pMMR/MSS cancer

Chemotherapy with one of the following regimens is recommended for unresectable pMMR/MSS cancers:

- FOLFIRI
- FOLFOX
- CAPEOX
- FOLFIRINOX

Bevacizumab (Avastin) may be added to any of these regimens.

Cancers that don't have *KRAS or NRAS* mutations are eligible for cetuximab (Erbitux) or panitumumab (Vectibix) in addition to chemotherapy. These are EGFR blockers. They may be added to the FOLFIRI or FOLFOX regimens.

This approach can also be considered for some *BRAF* mutations, but not *BRAF* V600E. EGFR blockers aren’t effective against cancers with this specific mutation unless a drug that blocks *BRAF* is also included.

While uncommon, systemic therapy may shrink the tumors enough to be removed or destroyed using surgery, ablation, or other local therapies. If this happens, your doctor may suggest treatment that includes radiation therapy before proceeding with surgery. If so, short-course radiation therapy is preferred, but chemoradiation is also a recommended option.

Sometimes chemotherapy and radiation work well enough that surgery isn’t needed, but this isn’t typical. If this happens, **see Guide 5 on page 29**.

If the tumors don't become resectable during chemotherapy, systemic therapy is typically continued. The goal is to slow the growth and spread of the cancer. If there aren't many liver or lung tumors, treatment with local therapies may also be an option.

If the rectal tumor grows during initial systemic therapy, you may first have chemoradiation or a short course of radiation to the rectal tumor before continuing systemic therapy.

If the cancer progresses, you may receive a different regimen next. One important factor is whether next-generation sequencing found any biomarkers. Regimens targeting specific biomarkers are listed in **Guide 6**.

If the cancer progresses through all recommended regimens, the following may be options:

- Targeted therapy with fruquintinib (Fruzaqla)
- Chemotherapy with trifluridine and tipiracil (Lonsurf) (bevacizumab may be added)
- Targeted therapy with regorafenib (Stivarga)

These therapies are tablets that can be taken by mouth.

Guide 6
Biomarker-based treatments for pMMR/MSS cancers

Biomarker	Targeted therapy options
***BRAF* V600E mutation**	• Encorafenib + (cetuximab or panitumumab) • Encorafenib + (cetuximab or panitumumab) + FOLFOX
HER2-amplified with normal *RAS* and *BRAF* genes	Trastuzumab (Herceptin) + pertuzumab, lapatinib, or tucatinib
HER2-amplified	Fam-trastuzumab deruxtecan-nxki (Enhertu)
***KRAS* G12C mutation**	• Sotorasib (Lumakras) + cetuximab or panitumumab • Adagrasib (Krasati) + cetuximab or panitumumab
***NTRK* gene fusion**	• Entrectinib (Rozlytrek) • Larotrectinib (Vitrakvi) • Repotrectinib (Augtyro)
***RET* gene fusion**	Selpercatinib (Retevmo)

dMMR/MSI-H or POLE/POLD1-mutated cancer

Treatment for stage 4 dMMR cancers starts with immunotherapy. Currently recommended checkpoint inhibitors are listed in **Guide 7**.

If you aren't eligible for immunotherapy (because you've already had it, for example), oxaliplatin-based chemotherapy is an option.

After immunotherapy, you will have imaging to see how it worked. Some people may need more treatment, which can include radiation therapy. Short-course radiation therapy is preferred, but chemoradiation is also a recommended option.

After immunotherapy and any radiation therapy, the rectal tumor and the liver or lung tumors are resected (surgically removed), often with other local therapies. The surgeries may be done at the same time, or as separate procedures.

Sometimes immunotherapy works so well that radiation and surgery aren't needed. In this case, surveillance will begin. Recommended surveillance for those treated without surgery is listed in **Guide 5 on page 29**.

Guide 7
Options for immunotherapy before surgery for dMMR/MSI-H cancers

Generic name	Brand name
nivolumab with or without ipilimumab	Opdivo, Yervoy
pembrolizumab	Keytruda
cemiplimab-rwlc	Libtayo
dostarlimab-gxly	Jemperli
retifanlimab-dlwr	Zynyz
toripalimab-pzi	Loqtorzi
tislelizumab-jsgr	Tevimbra

Stage 4 cancer in the abdomen

Some people with metastatic rectal cancer will also form tumors in the thin layer of tissue that lines the abdomen, called the peritoneum. The peritoneum covers most of the organs in the abdomen.

Most people receive treatment with systemic therapy. The goal is to relieve or prevent symptoms. Your care team will select a regimen based on whether the tumor has any biomarkers and your overall health.

If the tumor is blocking stool from moving and leaving the body, the bowel needs to be unblocked before starting systemic therapy. This is done using a surgical technique or with a mesh metal tube called a stent.

Surveillance

If you have a complete response to treatment that included surgery, the recommended surveillance schedule is shown in **Guide 8**.

In addition to surveillance testing, a range of other care is important for cancer survivors. For more information, see *Chapter 6: Survivorship*.

Guide 8
Surveillance after surgical treatment for stage 4 rectal cancer

• **Physical exam** • **CEA blood test** • **CT scan of your chest, abdomen, and pelvis**	**First 2 years:** Every 3 to 6 months **Next 3 years:** Every 6 months
Colonoscopy	**Did you have a complete colonoscopy at diagnosis?** If you **didn't**, your next colonoscopy is recommended in 3 to 6 months. If you **did**, you can wait until a year after surgery for your next colonoscopy.

Distant recurrence

After treatment for non-metastatic rectal cancer, the cancer may return and spread to the liver, lungs, or other areas. This is called a distant recurrence.

Treatment with surgery, ablation, or other local therapies is recommended if all of the tumors can be totally removed. But, this is rare, and most people receive treatment with systemic therapy.

While uncommon, systemic therapy may shrink the tumors enough to be removed with surgery or ablation. If the cancer doesn't become small enough to be resected or ablated, systemic therapy is typically continued. The goal is to slow the growth and spread of the cancer.

Treatment recommendations are provided next according to the mismatch repair status of the cancer and whether resection is possible.

Unresectable pMMR/MSS cancer

If you've had recent treatment with FOLFOX or CAPEOX, you should not have more chemotherapy that includes oxaliplatin. Oxaliplatin can cause serious nerve damage. Your options for systemic therapy will depend, in part, on whether the cancer has any of the biomarkers listed below. Therapies are available that target these biomarkers.

- HER2 amplification
- *KRAS* G12C mutation
- *BRAF* mutations
- *NTRK* gene fusion
- *RET* gene fusion

Supportive care

Supportive care helps improve your quality of life during and after cancer treatment. The goal is to prevent or manage side effects and symptoms, like pain and cancer-related fatigue. It also addresses the mental, social, and spiritual concerns faced by those with cancer.

Supportive care is available to everyone with cancer and their families, not just those at the end of life.

Supportive care can also help with:

- Making treatment decisions
- Coordinating your care
- Paying for care
- Planning for advanced care and end of life

For cancers without these biomarkers, chemotherapy with FOLFIRI or irinotecan is a recommended first-line option. Cancers that don't have *KRAS or NRAS* mutations are eligible for cetuximab (Erbitux) or panitumumab (Vectibix) in addition to chemotherapy. These are EGFR blockers. This approach can also be considered for some *BRAF* mutations, but not *BRAF* V600E. EGFR blockers aren't effective against cancers with

this specific mutation unless a drug that blocks *BRAF* is also included.

For cancers with *KRAS*, *NRAS*, or *BRAF* mutations, bevacizumab (Avastin), ziv-aflibercept (Zaltrap), or ramucirumab (Cyramza) may be added to chemotherapy.

Resectable pMMR/MSS cancer

For resectable pMMR/MSS cancers, treatment typically involves chemotherapy in addition to resection and treatment with local therapies.

Chemotherapy may be given before surgery, after surgery, or both.

If you **haven't** had any chemotherapy, treatment with resection and local therapies is often given first, followed by chemotherapy.

Unresectable dMMR/MSI-H or POLE/POLD1-mutated cancer

If you are a candidate and haven't had immunotherapy, treatment with a checkpoint inhibitor is recommended. At this time, recommended options include:

- Nivolumab (Opdivo) with or without ipilimumab (Yervoy)
- Pembrolizumab (Keytruda)
- Dostarlimab-gxly (Jemperli)

Your doctor will check the extent of the cancer every 2 to 3 months. Surgery may become possible. Or, you may continue immunotherapy or switch to a different systemic therapy.

If you've already had immunotherapy, chemotherapy is typically given. A biologic may be included in the regimen. Or, if the cancer has any other biomarkers, targeted therapy may be an option.

Resectable dMMR/MSI-H or POLE/POLD1-mutated cancer

If you haven't had any immunotherapy, recommended options include:

- Immunotherapy, followed by observation or resection
- Resection followed by chemotherapy

Local therapies may be used together with resection, or used alone for very small tumors.

At this time, recommended checkpoint inhibitors for treating distant recurrence include:

- Nivolumab (Opdivo) with or without ipilimumab (Yervoy)
- Pembrolizumab (Keytruda)
- Dostarlimab-gxly (Jemperli)

If you've had prior immunotherapy, one recommended option is resection first, followed by either chemotherapy or observation. Observation is recommended for those who have already had treatment with oxaliplatin.

A second option is 2 to 3 months of chemotherapy first, followed by resection. More chemotherapy may follow.

Clinical trials

A clinical trial is a type of medical research study. After being developed and tested in a lab, potential new ways of fighting cancer need to be studied in people.

If found to be safe and effective in a clinical trial, a drug, device, or treatment approach may be approved by the U.S. Food and Drug Administration (FDA).

Everyone with cancer should carefully consider all of the treatment options available for their cancer type, including standard treatments and clinical trials. Talk to your doctor about whether a clinical trial may make sense for you.

Phases

Most cancer clinical trials focus on treatment and are done in phases.

- Phase 1 trials study the safety and side effects of an investigational drug or treatment approach.
- Phase 2 trials study how well the drug or approach works against a specific type of cancer.
- Phase 3 trials test the drug or approach against a standard treatment. If the results are good, it may be approved by the FDA.
- Phase 4 trials study the safety and benefit of an FDA-approved treatment.

Who can enroll?

It depends on the clinical trial's rules, called eligibility criteria. The rules may be about age, cancer type and stage, treatment history, or general health. They ensure that participants are alike in specific ways and that the trial is as safe as possible for the participants.

Informed consent

Clinical trials are managed by a research team. This group of experts will review the study with you in detail, including its purpose and the risks and benefits of joining. All of this information is also provided in an informed consent form. Read the form carefully and

Finding a clinical trial

In the United States

NCCN Cancer Centers
NCCN.org/cancercenters

The National Cancer Institute (NCI)
cancer.gov/about-cancer/treatment/clinical-trials/search

Worldwide

The U.S. National Library of Medicine (NLM)
clinicaltrials.gov/

Need help finding a clinical trial?

NCI's Cancer Information Service (CIS)
1.800.4.CANCER (1.800.422.6237)
cancer.gov/contact

ask questions before signing it. Take time to discuss it with people you trust. Keep in mind that you can leave and seek treatment outside of the clinical trial at any time.

Will I get a placebo?

Placebos (inactive versions of real medicines) are almost never used alone in cancer clinical trials. It is common to receive either a placebo with a standard treatment, or a new drug with a standard treatment. You will be informed, verbally and in writing, if a placebo is part of a clinical trial before you enroll.

Are clinical trials free?

There is no fee to enroll in a clinical trial. The study sponsor pays for research-related costs, including the study drug. But you may need to pay for other services, like transportation or childcare, due to extra appointments. During the trial, you will continue to receive standard cancer care. This care is often covered by insurance.

Key points

- Everyone with metastatic rectal cancer should have biomarker tumor testing for *RAS* mutations, *BRAF* mutations, HER2 amplification, and mismatch repair deficiency (if not already performed).
- Next-generation sequencing (NGS) can find other, rare biomarkers for which targeted treatments may be available.
- Treatment options for metastatic rectal cancer may include local therapies, such as thermal ablation and SBRT.
- Stage 4 cancer that can't be removed with surgery and local therapies is treated with systemic therapy.
- For those who are eligible, checkpoint inhibitor immunotherapy is recommended for mismatch repair deficient (dMMR) and *POLE*/*POLD1*-mutated metastatic cancers.
- Supportive care is available for everyone with cancer to help with symptoms and make you more comfortable.
- Clinical trials provide access to investigational treatments that may, in time, be approved by the FDA.

Questions to ask

- Am I a candidate for a clinical trial? Do you know of one I can join?
- Am I a candidate for surgery? If not, is it possible that I'll become a candidate?
- Does my cancer have any biomarkers? How does this affect my options?
- Did an interventional oncologist/radiologist review my case, along with a multidisciplinary team, to see if I am a candidate for local therapies?
- Which systemic therapy regimen do you recommend for me? Why?

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Survivorship

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Survivorship begins on the day you learn you have rectal cancer. It focuses on the physical, emotional, and financial issues unique to cancer survivors.

Your primary care provider

After finishing cancer treatment, your oncologist and primary care provider should work together to provide recommended follow-up care. Ask your oncologist for a written survivorship care plan. Ideally, it would include:

- A summary of your cancer treatment history
- A description of possible short- and long-term side effects
- Recommendations for monitoring for recurrence
- Clear roles and responsibilities for your providers
- Recommendations on your overall health and well-being

Help with side effects

Rectal cancer survivors may experience both short- and long-term health effects of cancer and its treatment. The effects depend in part on the treatment(s) received. Surgery, anticancer medicines, and radiation therapy all have unique potential side effects.

Bowel changes

Surgery that involves removing all or part of the rectum is called a low anterior resection (LAR). This type of surgery often causes changes to how your bowel works, such as:

- Increase in bowel movements
- Urgent need to pass stool
- Having many (possibly watery) bowel movements over the course of a few hours (stool clustering)
- Irregular and unpredictable bowel movements
- Inability to control bowel movements (fecal incontinence)
- Increase in gas (flatulence)

Together these symptoms are referred to as **low anterior resection syndrome (LARS)**.

Loperamide (Imodium) is an over-the-counter anti-diarrheal medicine. It can help with stool clustering and incontinence. Exercises that strengthen the muscles in the pelvic floor can help with urgency and stool incontinence.

Making changes to your diet can also help with LARS symptoms. Eat foods high in insoluble (not soluble) fiber and use fiber supplements.

Psyllium-based products (like Metamucil) may help slow and thicken the stool.

Ask your care team for a list of foods that may help or worsen bowel changes after low anterior resection. Many people find that protective undergarments provide peace of mind. A cancer nutritionist can also help you make changes to your diet after rectal surgery.

Nerve damage

The chemotherapy drug oxaliplatin can damage nerves in your fingers and toes. Symptoms include numbness, cramping, tingling, or pain in these areas. Acupuncture and heat may help. If you have painful nerve damage, a drug called duloxetine (Cymbalta) may help. If the pain is persistent, talk to your doctor about seeing a pain management specialist.

Ostomy care

If you have an ostomy, you may want to join an ostomy support group. Another option is to see a health care provider that specializes in ostomy care, such as an ostomy nurse.

People with ostomies can still live very active lifestyles. However, it's a good idea to talk to an ostomy professional before doing any intense physical activity.

Sexual health

Many people experience sexual side effects after chemotherapy and radiation. For those with a penis, this could mean difficulty getting an erection. Those with a vagina may experience vaginal dryness and pain during sex.

Radiation therapy to the pelvis can cause the vagina to become shorter and more narrow. This is called vaginal stenosis.

Ostomy care

You can still live an active lifestyle with an ostomy. Joining a support group or seeing an ostomy nurse can help you learn more about living with an ostomy.

Vaginal stenosis can make it uncomfortable or even painful to have sex, or to have vaginal examinations by a doctor. Vaginal dilator therapy can help. A vaginal dilator is a device used to gradually stretch or widen the vagina. Vaginal dilators are not one-size-fits-all.

Different sizes are available, as are kits containing different size devices. The size of the dilator can be increased over time as the vagina lengthens and widens. Use of a vaginal dilator can begin as soon as you are healed after treatment.

Menopausal hormone therapy

The ovaries make hormones needed for menstruation and natural pregnancy. Radiation damages the ovaries, causing hormone levels to drop. This is known as premature ovarian insufficiency (POI) or radiation-induced premature menopause. It causes infertility and may also cause symptoms of menopause, including:

- Stopping of periods
- Hot flashes
- Sleeping problems
- Night sweats
- Weight gain
- Changes in mood
- Vaginal dryness

Menopausal hormone therapy may help lessen these side effects. This approach used to be called hormone replacement therapy or HRT.

One option is systemic combination therapy. Both estrogen and progestogen are given, usually either as a pill or a patch placed on the skin. The hormones travel throughout the body in the bloodstream. Progestogen is included because giving estrogen by itself can increase the risk of uterine cancer in those with a uterus.

Another option for menopausal hormone therapy is vaginal estrogen cream or tablets. This type may be the best option for symptoms that mainly affect the vagina, such as dryness.

Discussion with a specialized menopausal symptom team may be helpful to determine whether this treatment is right for you.

Healthy habits

It's important to keep up with other aspects of your health after cancer treatment. In addition to the recommendations on the next page, ask your provider about the following.

Low GL diet

A low sugar (called glycemic load or GL) diet may help prevent the return of rectal cancer. Low GL foods cause a slower and smaller rise in blood sugar levels compared to other carbohydrate-containing foods.

Aspirin

Talk to your doctor about the possible risks and benefits of long-term aspirin therapy to help prevent the return of colorectal cancers.

Healthy habits for cancer survivors

Follow your primary care provider's recommendations for screening for other cancers based on your age, sex, and risk level.

Get other recommended health care for your age, such as blood pressure checks, hepatitis C screening, and immunizations (like the flu shot)

Eat a diet rich in plant-based foods. Limit red meat and processed foods. Ask your provider for a referral to a cancer nutritionist.

Alcohol can increase the risk of some cancers. Drink little to no alcohol.

Move more, rest less. Try to exercise at a moderate intensity for at least 150 minutes per week.

Maintain a healthy weight. Tracking your weight, diet, calories, and activity levels may help you meet your goals.

Stop using tobacco. Ask your care team about options to help you quit. Counseling and medication are often options.

Paying for care

Cancer survivors face a unique financial burden. Paying for doctor visits, tests, and treatments can become unmanageable, especially for those with little or no health insurance. You may also have costs not directly related to treatment, such as travel expenses and the cost of childcare.

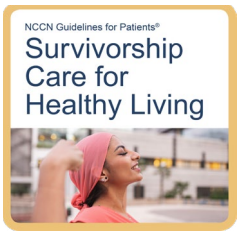
The term financial toxicity is used to describe the problems patients face related to the cost of medical care. Financial toxicity can affect your quality of life and access to needed health care.

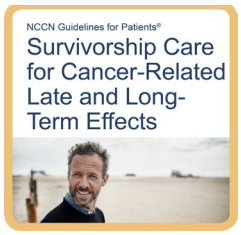
If you need help paying for cancer care, financial assistance may be available. Talk with a patient navigator, your care team's social worker, and your hospital's financial services department.

More information

For more information on cancer survivorship, the following are available at NCCN.org/patientguidelines:

- *Survivorship Care for Healthy Living*
- *Survivorship Care for Cancer-Related Late and Long-Term Effects*





Key points

- Survivorship focuses on the physical, emotional, and financial issues unique to cancer survivors.
- Ideally, your oncologist and primary care provider will work together to provide recommended follow-up care. Nurse navigators can also be helpful resources.
- A survivorship care plan is helpful in transitioning your care to your primary care provider.
- Healthy habits, including exercising and eating right, play an important role in helping to prevent other diseases and second cancers.

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Other resources

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Want to learn more? Here's how you can get additional help.

What else to know

This book can help you improve your cancer care. It plainly explains expert recommendations and suggests questions to ask your care team. But it's not the only resource that you have.

You're welcome to receive as much information and help as you need. Many people are interested in learning more about:

- Details of their care and treatment
- Getting financial help
- Finding a care provider who is an expert in their field
- Coping with health effects

What else to do

Your health care center can help you with next steps. They often have on-site resources to help meet your needs and find answers to your questions. Health care centers can also inform you of resources in your community.

In addition to help from your providers, the resources listed in the next section provide support for many people like yourself. Look through the list and visit the provided websites to learn more about these organizations.

Where to get help

Anal Cancer Foundation
analcancerfoundation.org

Bone Marrow and Cancer Foundation
bonemarrow.org

CancerCare
cancercare.org

Cancer Hope Network
cancerhopenetwork.org

Cancer Survivor Care
cancersurvivorcare.org

Colorectal Cancer Alliance
ccalliance.org

Fight Colorectal Cancer
fightcolorectalcancer.org

GRACE
cancergrace.org

HPV Cancers Alliance
hpvalliance.org

Imerman Angels
Imermanangels.org

Love Your Buns
loveyourbuns.org

My Faulty Gene
myfaultygene.org

National Coalition for Cancer Survivorship
canceradvocacy.org

Paltown Development Foundation
paltown.org

PAN Foundation
panfoundation.org

Triage Cancer
triagecancer.org

Let us know what you think!

Please take a moment to complete an online survey about the NCCN Guidelines for Patients.
NCCN.org/patients/response

Words to know

ablation
A type of local therapy used to destroy tumors in the liver or lungs. Also called image-guided thermal ablation.

adenocarcinoma
Cancer in cells that line organs and make fluids or hormones. The most common type of rectal cancer.

adenoma
The most common type of colorectal polyp and the most likely to form cancer cells. Also called adenomatous polyp.

biomarkers
Specific features of cancer cells used to guide treatment. Biomarkers are often mutations (changes) in the DNA of the cancer cells.

CAPEOX
A chemotherapy regimen that includes capecitabine and oxaliplatin.

carcinoembryonic antigen (CEA)
A protein that gets released by some tumors and can be detected in blood.

chemoembolization
Treatment for metastatic liver tumors that delivers spheres carrying chemotherapy to the liver vessels supplying the tumor.

colon
The first and longest section of the large bowel. Unused food is turned into stool in the colon.

colostomy
Surgery to connect a part of the colon to the outside of the abdomen and that allows stool to drain into a bag.

endoscopic submucosal dissection (ESD)
A minimally invasive type of surgery used to remove small rectal tumors in one piece. Also called endoscopic mucosal resection (EMR).

external beam radiation therapy (EBRT)
Treatment with high-energy rays received from a machine outside the body.

familial adenomatous polyposis (FAP)
An inherited cancer syndrome that increases the risk of colorectal cancer.

FOLFIRI
A chemotherapy regimen used for some advanced colorectal cancers. Includes leucovorin calcium, fluorouracil, and irinotecan.

FOLFIRINOX
A chemotherapy regimen used for some advanced colorectal cancers. Includes leucovorin calcium (folinic acid), fluorouracil, irinotecan, and oxaliplatin.

FOLFOX
A chemotherapy regimen that includes leucovorin calcium, fluorouracil, and oxaliplatin.

interventional oncology/radiology
A medical specialty that uses imaging techniques to deliver minimally invasive cancer treatments.

irreversible electroporation
A type of local therapy that uses electrical pulses to destroy small tumors.

large intestine (bowel)
A long tube-shaped organ that forms the last part of the digestive system. Includes the colon, rectum, and anus.

lymph
A clear fluid containing white blood cells.

lymph node
Small groups of special disease-fighting cells located throughout the body.

metastasis
The spread of cancer cells from the first (primary) tumor to a distant site.

mismatch repair deficiency (dMMR)/high microsatellite instability (MSI-H)
A biomarker (feature) of some colorectal cancers that is used to guide treatment. All colorectal cancers should be tested for this biomarker.

mismatch repair proficient (pMMR)/ microsatellite stable (MSS)
Describes cancers that don't have the mismatch repair deficiency (dMMR) biomarker or the microsatellite instability-high (MSI-H) biomarker.

pathologist
A doctor who specializes in testing cells and tissue to find disease.

polyp
An overgrowth of cells on the inner lining of the colon wall. Pedunculated polyps are shaped like mushrooms with a stalk. Sessile polyps are flat.

portal vein embolization
The blood vessel to the liver tumor is blocked causing the healthy part of the liver to grow larger.

radioembolization
A treatment for liver tumors that uses radioactive spheres injected in the hepatic (liver) artery (the blood vessel supplying tumor). Also called selective internal radiation therapy or SIRT.

rectum
An organ in the digestive system that holds stool until it exits the body through the anus.

recurrence
The return of cancer after a cancer-free period.

resectable
Describes a cancer that can be safely removed using surgery.

stereotactic body radiation therapy (SBRT)
A highly specialized type of radiation therapy. May be used to treat rectal cancer that has spread to the liver, lungs, or bone.

supportive care
Treatment for the symptoms or health conditions caused by cancer or cancer treatment.

surgical margin
The normal tissue around the edge of a tumor removed during surgery.

systemic therapy
The use of medicines that enter the bloodstream. Chemotherapy, targeted therapy, and immunotherapy are examples.

unresectable
Describes a cancer that can't be safely removed using surgery.

NCCN Contributors

This patient guide is based on the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) forRectal Cancer, Version 3.2025 — August 26, 2025. It was adapted, reviewed, and published with help from the following people:

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NCCN Cancer Centers

Abramson Cancer Center
at the University of Pennsylvania
Philadelphia, Pennsylvania
800.789.7366 • pennmedicine.org/cancer

Case Comprehensive Cancer Center/
University Hospitals Seidman Cancer Center and
Cleveland Clinic Taussig Cancer Institute
Cleveland, Ohio
UH Seidman Cancer Center
800.641.2422 • uhhospitals.org/services/cancer-services
CC Taussig Cancer Institute
866.223.8100 • my.clevelandclinic.org/departments/cancer
Case CCC
216.844.8797 • case.edu/cancer

City of Hope National Medical Center
Duarte, California
800.826.4673 • cityofhope.org

Dana-Farber/Brigham and Women's Cancer Center |
Mass General Cancer Center
Boston, Massachusetts
877.442.3324 • youhaveus.org
617.726.5130 • massgeneral.org/cancer-center

Duke Cancer Institute
Durham, North Carolina
888.275.3853 • dukecancerinstitute.org

Fox Chase Cancer Center
Philadelphia, Pennsylvania
888.369.2427 • foxchase.org

Fred & Pamela Buffett Cancer Center
Omaha, Nebraska
402.559.5600 • unmc.edu/cancercenter

Fred Hutchinson Cancer Center
Seattle, Washington
206.667.5000 • fredhutch.org

Huntsman Cancer Institute at the University of Utah
Salt Lake City, Utah
800.824.2073 • healthcare.utah.edu/huntsmancancerinstitute

Indiana University Melvin and Bren Simon
Comprehensive Cancer Center
Indianapolis, Indiana
888.600.4822 • www.cancer.iu.edu

Johns Hopkins Kimmel Cancer Center
Baltimore, Maryland
410.955.8964
www.hopkinskimmelcancercenter.org

Mayo Clinic Comprehensive Cancer Center
Phoenix/Scottsdale, Arizona
Jacksonville, Florida
Rochester, Minnesota
480.301.8000 • Arizona
904.953.0853 • Florida
507.538.3270 • Minnesota
mayoclinic.org/cancercenter

Memorial Sloan Kettering Cancer Center
New York, New York
800.525.2225 • mskcc.org

Moffitt Cancer Center
Tampa, Florida
888.663.3488 • moffitt.org

O'Neal Comprehensive Cancer Center at UAB
Birmingham, Alabama
800.822.0933 • uab.edu/onealcancercenter

Robert H. Lurie Comprehensive Cancer Center
of Northwestern University
Chicago, Illinois
866.587.4322 • cancer.northwestern.edu

Roswell Park Comprehensive Cancer Center
Buffalo, New York
877.275.7724 • roswellpark.org

Siteman Cancer Center at Barnes-Jewish Hospital
and Washington University School of Medicine
St. Louis, Missouri
800.600.3606 • siteman.wustl.edu

St. Jude Children's Research Hospital/
The University of Tennessee Health Science Center
Memphis, Tennessee
866.278.5833 • stjude.org
901.448.5500 • uthsc.edu

Stanford Cancer Institute
Stanford, California
877.668.7535 • cancer.stanford.edu

The Ohio State University Comprehensive Cancer Center -
James Cancer Hospital and Solove Research Institute
Columbus, Ohio
800.293.5066 • cancer.osu.edu

The UChicago Medicine Comprehensive Cancer Center
Chicago, Illinois
773.702.1000 • uchicagomedicine.org/cancer

The University of Texas MD Anderson Cancer Center
Houston, Texas
844.269.5922 • mdanderson.org

UC Davis Comprehensive Cancer Center
Sacramento, California
916.734.5959 • 800.770.9261
health.ucdavis.edu/cancer

UC San Diego Moores Cancer Center
La Jolla, California
858.822.6100 • cancer.ucsd.edu

UCLA Jonsson Comprehensive Cancer Center
Los Angeles, California
310.825.5268 • uclahealth.org/cancer

UCSF Helen Diller Family Comprehensive Cancer Center
San Francisco, California
800.689.8273 • cancer.ucsf.edu

University of Colorado Cancer Center
Aurora, Colorado
720.848.0300 • coloradocancercenter.org

University of Michigan Rogel Cancer Center
Ann Arbor, Michigan
800.865.1125 • rogelcancercenter.org

University of Wisconsin Carbone Cancer Center
Madison, Wisconsin
608.265.1700 • uwhealth.org/cancer

UT Southwestern Simmons Comprehensive Cancer Center
Dallas, Texas
214.648.3111 • utsouthwestern.edu/simmons

Vanderbilt-Ingram Cancer Center
Nashville, Tennessee
877.936.8422 • vicc.org

Yale Cancer Center/Smilow Cancer Hospital
New Haven, Connecticut
855.4.SMILOW • yalecancercenter.org

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NCCN.org/patients/comments

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